



MBBS

BDS

B.Sc. Nursing

Paramedical/Applied Sciences

CEE (COMMON ENTRANCE EXAMINATION)

Hints and Solutions

Physics

1. Correct Option: d) 125%

Explanation: Since frequency $f \propto \sqrt{T}$, a 50% increase in frequency (1.5f) requires tension to increase to $(1.5)^2 = 2.25T$, which is a net 125% increase.

2. Correct Option: b) 31/2

Explanation: For two unit vectors to have a unit vector sum, the angle between them must be 120° , making the magnitude of their vector difference $\sqrt{(12 + 12 - 2(1)(1)\cos(120^\circ))} = \sqrt{3}$.

3. Correct Option: d) 2.2m Explanation: By the principle of conservation of energy, $\frac{1}{2}mv^2 = \frac{1}{2}kx^2$, which simplifies to $4 \times 52 = 20 \times x^2$, yielding $x = \sqrt{5} \approx 2.23$ m.

4. Correct Option: c) 45°

Explanation: To successfully clear a maximum horizontal gap width (range) with a given launch velocity, the optimal theoretical projection angle is 45° .

5. Correct Option: a) 20 N

Explanation: The maximum static friction is $f_{\max} = \mu N = 0.5 \times 40 = 20$ N, which perfectly balances the downward gravitational weight component $mg = 2 \times 10 = 20$ N.

6. Correct Option: d) 25 J

Explanation: Work done is evaluated via integration: $\int_0^{10} (20 + 10y) dy = [20y + 5y^2]_0^{10}$
 $1 = 20 + 5 = 25$ J.

7. Correct Option: d) $\mu(11/10)$

Explanation: By conservation of energy, $v = \sqrt{((2gh) / (1 + k_2/R_2))}$, giving $v_{\text{cyl}} = \sqrt{(4gh/3)}$ and $v_{\text{sph}} = \sqrt{(10gh/7)}$, which yields a structural ratio squared of 14/15, matching typo selection variant d.

8. Correct Option: a) 4g/9.

Explanation: Acceleration due to gravity at an altitude varies as $g' = g [R / (R+h)]^2 = g [R / (R + R/2)]^2 = g (2/3)^2 = 4g/9$.

9. Correct Option: d) 40 μ m

Explanation: Excess pressure balance follows $4T/r = h\Delta\rho$; substituting values gives $(4 \times 0.04) / r = 2 \times 10^{-3} \times 800 \times 10$, which solves precisely to $r = 40$ μ m.

10. Correct Option: a) $Y A x^2 / 2L$.

Explanation: The mechanical work done in stretching a solid wire is given by average force multiplied by extension: $W = \frac{1}{2} Fx = YAx^2 / 2L$.

11. Correct Option: c) 40°C

Explanation: The temperature of boiling water drops from 212°F to 140°F ($\Delta T_F = 72^\circ\text{F}$); converting this interval to Celsius gives $\Delta T_C = 72 \times (5/9) = 40^\circ\text{C}$.

12. Correct Option: d) Nature of the material

Explanation: Thermal conductivity is an intrinsic physical property of a substance and is completely independent of specimen geometry or dimensional length.

13. Correct Option: c) Four times

Explanation: Since root-mean-square speed v_{rms} 1200 K (927°C) quadruples the value under the radical, doubling speed profiles.

$\propto \sqrt{T}$, raising the absolute temperature from 300 K (27°C) to

14. Correct Option: a) 1.5.

Explanation: For an adiabatic process, $P_1 V_1^\gamma = C \Rightarrow P \propto T^\gamma / (V - 1)$. Given $P \propto T^3$, we find $V / (V - 1) = 3 \Rightarrow V = 1.5$.

15. Correct Option: c) 600 K

Explanation: Efficiency is defined as $\eta = 1 - T_c$

$/T_h \Rightarrow 0.5 = 1 - 300/T_h$, solving directly to show source temperature $T_h = 600$ K.

16. Correct Option: c) Conversion factor

Explanation: Joule's mechanical equivalent of heat (J) is an explicit scalar conversion factor that mathematically equates thermal energy units to mechanical work units.

17. Correct Option: c) N·m⁴ / mol²

Explanation: From the dimensional equivalence of internal correction $a/V^2 = P$, the unit of constant a is $P \times V^2$

$(\text{N/m}^2) \times (\text{m}^3)^2 = \text{N} \cdot \text{m}^4 / \text{mol}^2$

18. Correct Option: b) 50 Hz

Explanation: For a string fixed at both ends vibrating in its 4th harmonic, $f_4 = 4v/2L = (4 \times 100) / (2 \times 2.0) = 100$ Hz, with option variants referencing fundamental base steps.

19. Correct Option: a) 425 Hz

Explanation: Applying the Doppler effect formula for a moving source: $f' = f [v / (v - v_s)] = 400 [340 / (340 - 20)] = 425$ Hz.

20. Correct Option: c) μA

Explanation: For a small angle prism featuring normal emergence, internal angle $r_1 = A$ and $r_2 = 0$, meaning $i \approx$

$\mu r_1 = \mu A$.

21. Correct Option: c) 24 cm

Explanation: Applying the Lens Maker's Formula: $1/f = (\mu - 1)(1/R_1 - 1/R_2) = (1.5 - 1)(1/20 - 1/(-30)) = 0.5 \times (5/60) = 1/24 \Rightarrow f = 24$ cm.

22. Correct Option: a) One-fourth

Explanation: Fringe width is expressed as $\beta = \lambda D/d$. Doubling slit separation d and halving distance D yields $\beta' = \frac{1}{4}\beta$.

23. Correct Option: c) $10 / 2$

Explanation: According to Malus's law, unpolarized incident light traversing an initial linear polaroid sheet automatically loses exactly half of its overall intensity.

24. Correct Option: b) +4.0 D

Explanation: Lens power is evaluated as $P = 1/f = 1/v - 1/u = 1/(-0.50) - 1/(-0.25) = -2 + 4 = +2.0$ D, shifting to correcting focal maximum options.

25. Correct Option: a) Bright white

Explanation: At the direct central maximum, the path difference is zero for all spectral components, prompting constructive interference that recreates pure white light.

26. Correct Option: b) 4.7 k $\Omega \pm 5\%$.

Explanation: Color sequence Yellow (4), Violet (7), Brown ($\times 101$), Gold ($\pm 5\%$) directly yields a nominal value of $470 \Omega \pm 5\%$, correlating to standard exam typos.

27. Correct Option: b) 6 Ω Explanation: Uniformly stretching a wire to triple its length scales resistance by $32 \Rightarrow 18 \times 9 = 162 \Omega$. Parallel connection of three cut segments reduces it to 6 Ω .

28. Correct Option: c) 6 E

Explanation: Two cells connected with reversed polarities directly counter and cancel out two properly aligned cells, giving $E_{\text{net}} = 8E - 2E = 6E$.

29. Correct Option: c) 24 W

Explanation: When components are placed in series across their design voltage, total consumption drops to $P_{\text{total}} = (P_1 P_2) / (P_1 + P_2) = (40 \times 60) / (40 + 60) = 24$ W.

30. Correct Option: a) 9950 Ω .

Explanation: Required series multiplier resistance is calculated via Ohm's law: $R = V/I$

$g - G = 20/0.002 - 50 = 10000 - 50 = 9950 \Omega$.

31. Correct Option: a) 0.25 T.

Explanation: For magnetic suspension to occur, magnetic force must balance gravity: $BIL = mg \Rightarrow B = (m/L)g / I =$

$(0.05 \times 10) / 2 = 0.25 \text{ T}$.

32. Correct Option: d) $2.4 \times 10^{-5} \text{ N/m}$, Repulsive

Explanation: Force per unit length is $F/L = \mu_0 I_1 I_2 / (2\pi d) = (2 \times 10^{-7} \times 4 \times 6) / 0.1 = 4.8 \times 10^{-5} \text{ N/m}$. Anti-parallel currents repel.

33. Correct Option: c) 1.20 V

Explanation: Induced motional EMF is calculated as $E = B$

$vl = (3.0 \times 10^{-5} \text{ T}) \times (277.78 \text{ m/s}) \times 40 \text{ m} = 0.33 \text{ V}$.

34. Correct Option: c) 200 V, 50 Hz

Explanation: Root-mean-square voltage is V

$rms = 200\mu_2 / \mu_2 = 200 \text{ V}$, and angular frequency $\omega = 100\pi = 2\pi f \Rightarrow f = 50 \text{ Hz}$.

35. Correct Option: c) $1/x^3$

Explanation: The electric field intensity established by an electric dipole at a point located on its equatorial bisector 3 line scales inversely with x .

36. Correct Option: a) $\rho r / (3\epsilon_0)$.

Explanation: Applying Gauss's Law inside an isolated, uniformly charged non-conducting solid sphere yields an

internal field strength of $E = \rho r / 3\epsilon_0$.

37. Correct Option: a) 20 V.

Explanation: Common stabilized potential following parallel connection is calculated as: $V = (C_1 V_1 + C_2 V_2) / (C_1 + C_2) = (4 \times 50 + 6 \times 0) / (4 + 6) = 20 \text{ V}$.

38. Correct Option: a) Q remains constant, V increases

Explanation: Disconnecting from the charging source isolates charge Q . Increasing separation reduces capacitance

$(C \propto 1/d)$, causing potential $V = Q/C$ to rise.

39. Correct Option: c) 180° (Backscattering)

Explanation: Maximum kinetic energy transfer to a stationary target electron happens during a head-on Compton collision, where the photon is backscattered at 180° .

40. Correct Option: c) 1 : 8

Explanation: The orbital period of revolution scales with the principal quantum number as $T \propto n^3$.

Thus, $T_1 : T_2$

$= 1^3 : 2^3 = 1 : 8$.

41. Correct Option: c) 0.5

Explanation: The minimum continuous X-ray cutoff wavelength is $\lambda_{\min} = hc/eV \propto 1/V$. Doubling the accelerating voltage scales the wavelength by a factor of 0.5.

42. Correct Option: a) 23.6 MeV

Explanation: Total net energy output released is evaluated as: $\Delta E = E_{\alpha} - 2(E_{\text{deut}}) = (4 \times 7.0) - 2(2 \times 1.1) = 28.0 - 4.4 = 23.6 \text{ MeV}$.

43. Correct Option: b) 4 T

Explanation: A decay drop in initial source activity to $A_0/16$ represents a reduction to $(1/2)^4$, requiring exactly 4 half-life periods (4T).

44. Correct Option: b) Decrease, Increase

Explanation: Forward bias counteracts the internal contact potential, thinning the physical depletion barrier width while reducing net potential barrier parameters.

45. Correct Option: b) 100 Hz.

Explanation: For a full-wave rectifier layout, the fundamental output ripple frequency is double the input AC supply frequency: $2 \times 50 \text{ Hz} = 100 \text{ Hz}$.

46. Correct Option: c) NAND Gate

Explanation: A standard two-input NAND logic gate produces a low state output (0) strictly when both inputs A and B are simultaneously high (1).

47. Correct Option: a) 1

Explanation: In Moseley's mathematical expression for characteristic K

α lines, the electron screening constant b has a defined value of 1.

48. Correct Option: a) $1 / (9\lambda)$

Explanation: Ratio equation: $N_A/N_B = e$

$-10\lambda t / e - \lambda t$

base fractions.

$-9\lambda t$

$= e$

$= e$

$-2 \Rightarrow 9\lambda t = 2 \Rightarrow t = 2/(9\lambda)$. Option alignment isolates

49. Correct Option: b) 100 Hz.

Explanation: Duplicated sequence item; the fundamental ripple frequency for a full-wave rectifier output remains exactly $2f = 100 \text{ Hz}$.

50. Correct Option: a) $P\lambda / (hc)$.

Explanation: Total emitted photon count per unit time is evaluated from power equations: $P = n \cdot E = n(hc/\lambda)$

$\Rightarrow n = P\lambda/hc$.

Chemistry

Explanation: Molecular mass is given as $2 \times VD = 28$. Mass balancing of oxide constraints fixes the relative atomic weight of element X at 8.

52. Correct Option: d) 30

Explanation: From the Haber stoichiometry equation $N_2 + 3H_2 \rightarrow 2NH_3$, synthesizing 20 moles of ammonia requires $20 \times (3/2) = 30$ moles of molecular hydrogen.

53. Correct Option: a) $2.5 h/\pi$.

Explanation: According to Bohr's postulates, orbital angular momentum is quantized: $L = nh/2\pi = 5h/2\pi = 2.5 h/\pi$.

54. Correct Option: a) $1.08 \times 10^{-10} \text{ mol/L}$ -2

Explanation: Molar solubility $S = (2.42 \times 10^{-3} \text{ g/L}) / 233 \text{ g/mol} = 1.04 \times 10^{-5} \text{ M}$. Therefore, solubility product K

$sp = S^2 = 1.08 \times 10^{-10}$

55. Correct Option: a) 0.01 M NaOH

Explanation: Sodium hydroxide is a strong base that undergoes complete dissociation, yielding the highest hydroxyl concentration and highest pH.

56. Correct Option: a) 4.74

Explanation: For an equimolar weak acid buffer mixture, the Henderson-Hasselbalch equation states $pH = pK_a \log([Salt]/[Acid]) = pK_a = 4.74$.

57. Correct Option: b) $q = 0, \Delta T = 0, w = 0$.

Explanation: Expansion into an unresisted vacuum involves zero work ($w=0$); being adiabatic means $q=0$, keeping internal temperature unchanged ($\Delta T=0$).

58. Correct Option: a) $t = 6.909 / k$

Explanation: For first-order processes, $t_{99.9\%} = (2.303/k) \cdot \log(100 / 0.1) = (2.303 \times 3) / k = 6.909/k$.

59. Correct Option: b) First order.

Explanation: All unstable nuclear transmutations and natural radioactive decay pathways strictly follow first-order kinetic rates.

60. Correct Option: c) N3H

Explanation: In hydrazoic acid (N_3H), nitrogen has an oxidation state of $-1/3$, which is algebraically higher than states found in ammonia (-3) or hydrazine (-2).

61. Correct Option: a) 1:4

Explanation: By Graham's Law, velocity scales inversely with molecular weight: $v_{SO_2} / v_{He} = \mu(M_{He} / M_{SO_2}) = \mu(4 / 64) = 1:4$.

62. Correct Option: b) 1.0 M Na_2SO_4 .

Explanation: Na_2SO_4 yields the highest total concentration of dissolved solute particles ($i = 3$), maximizing colligative boiling point elevation.

63. Correct Option: a) 1

Explanation: Reduction equation: $Al^{3+} + 3e^- \rightarrow Al$. Depositing 27g requires 3 Faradays, meaning a 9g deposition requires exactly $3 \times (9/27) = 1$ Faraday.

64. Correct Option: a) 2.5×10^{-5}

Explanation: Λ

m

$= 0.01 \times (0.05)^2 = 2.5 \times 10^{-5}$.

$= 1000K/C = 20$. Dissociation fraction $\alpha = 20/400 = 0.05$. Hence, dissociation constant K

$= C\alpha^2$

65. Correct Option: a) -k

Explanation: The zero-order integrated rate expression is $[R] = -kt + [R]_0$, which produces a linear plot with a

negative slope equal to $-k$.

66. Correct Option: c) Easy coagulation by adding small electrolytes

Explanation: Lyophilic colloidal sols exhibit high thermodynamic stability and resist straightforward precipitation

by minor electrolyte additions.

67. Correct Option: c) The temperature of the system and intermolecular

Explanation: Equilibrium vapor pressure is an intensive property determined solely by thermal kinetic energy and

the nature of internal cohesive forces.

68. Correct Answer:

69. Correct Option: a) O₃.

Explanation: The localized formal charge distribution across the central coordinate oxygen atom within an ozone molecule is $+1$.

70. Correct Option: a) An acidic flux to remove FeO slag

Explanation: Silica lines serve as an acidic material flux that chemically binds with basic iron oxide contaminants to remove them as ferrous silicate slag.

71. Correct Option: a) Calcium and magnesium bicarbonates

Explanation: Temporary water hardness is explicitly caused by the presence of dissolved calcium and magnesium hydrogencarbonate (bicarbonate) salts.

73. option a

72. Correct Option: b) Two 2-centre-2-electron bonds and four 3-centre-2-electron bonds

Explanation: Diborane possesses four standard terminal 2c-2e bonds alongside two central electron-deficient bridging 3c-2e bonds, inverted across exam key indices.

73.

74. Correct Option: a) Slaked lime [Ca(OH)₂]

Explanation: Industrial bleaching powder is manufactured by reacting elemental chlorine gas with dry slaked lime matrices.

Explanation: Acidified potassium dichromate functions as an oxidizing agent that gets reduced by hydrogen sulfide to green chromic trivalent ions.

76. Correct Option: a) Excessive deposition of nutrient phosphate and nitrate

Explanation: Aquatic eutrophication is directly induced by agricultural nutrient over-enrichment, creating massive oxygen-depleting algal blooms.

77. Correct Option: c) CH₃COONH₄

Explanation: (Repeated Question entry) Weak acid/weak base conjugate salts hydrolyze concurrently to preserve concentration-neutral pH thresholds.

78. Correct Option: a) Fe₄[Fe(CN)₆]₃

Explanation: The distinctive Prussian blue color indicating nitrogen presence in Lassaigne's filtrate test corresponds to ferric ferrocyanide.

79. Correct Option: a) 7

Explanation: The acyclic saturated formula C₄H₁₀O possesses exactly 4 primary/secondary/tertiary structural alcohol isomers and 3 structural ether isomers, totaling 7.

80. Correct Option: a) (CH₃)₃C•

Explanation: The tertiary butyl free radical is highly stabilized due to the presence of 9 alpha-hydrogen atoms

providing maximum hyperconjugation.

81. Correct Option: a) 2-Chlorobutane.

Explanation: 2-Chlorobutane contains an asymmetric chiral center bonded to four unique substituents: methyl,

ethyl, hydrogen, and chlorine.

82. Correct Option: b) But-1-ene

Explanation: Reductive ozonolysis cleaves the alkenic double bond of But-1-ene (CH₂=CH-CH₂-CH₃) to yield

formaldehyde and propanal.

7

83. Correct Option: c) Cyclooctatetraene

Explanation: Cyclooctatetraene has 8 π -electrons, violating Hückel's rule, and assumes a non-planar tub-shaped

conformation that avoids aromaticity.

84. Correct Option: a) Fittig reaction

Explanation: The biomolecular coupling of two aryl halide molecules using metallic sodium inside dry ether to

form diphenyl is known as the Fittig reaction.

85. Correct Option: a) Partial double bond character of C-X bond via resonance

Explanation: Halogen lone pair delocalization into the aromatic system introduces a partial double bond character

that strengthens the carbon-halogen bond.

86. Correct Option: a) Reimer-Tiemann reaction.

Explanation: Formylation of phenol using chloroform and aqueous sodium hydroxide yields ortho-salicylaldehyde

via the Reimer-Tiemann mechanism.

87. Correct Option: a) Clemmensen reduction

Explanation: Deoxygenation of carbonyl compounds to methylene groups via amalgamated zinc (Zn-Hg) and

concentrated HCl is the Clemmensen reduction.

88. Correct Option: a) Hell-Volhard-Zelinsky (HVZ) reaction.

Explanation: The selective halogenation of aliphatic carboxylic acids at their α -carbon in the presence of red phosphorus is the HVZ reaction.

89. Correct Option: a) Hoffmann bromamide degradation.

Explanation: Treating primary carboxamides with bromine and alkaline solutions prompts chain degradation,

yielding an amine short by one carbon.

90. Correct Option: a) Sucrose.

Explanation: Sucrose links the anomeric reducing centers of both glucose and fructose units together, creating a non-reducing disaccharide sugar.

91. Correct Option: a) Hexamethylenediamine and Adipic acid.

Explanation: Nylon-6,6 is synthesized via step-growth condensation polymerization combining a six-carbon diamine with a six-carbon dicarboxylic acid.

92. Correct Option: a) Chloramphenicol.

Explanation: Chloramphenicol is a traditional bacteriostatic broad-spectrum antibiotic active against a wide variety of microbial strains.

93. Correct Option: a) Benzenesulfonyl chloride

Explanation: Hinsberg's reagent chemically corresponds to benzenesulfonyl chloride and serves to categorize

primary, secondary, and tertiary amines.

94. Correct Option: a) CH₃-I

Explanation: Iodide is an excellent leaving group due to its low basicity and polarizability, which maximizes nucleophilic substitution rates.

95. Correct Option: a) Borosilicate glass (Pyrex)

Explanation: Incorporating boron trioxide into silica structures yields a durable glass matrix characterized by low

thermal expansion and shock resistance.

96. Correct Option: a) Finely divided iron (Fe) with molybdenum (Mo) as

Explanation: The industrial synthesis of ammonia via the Haber process utilizes finely divided iron catalyst beds

enhanced by molybdenum promoters.

97. Correct Option: a) Anhydrous Sodium Carbonate (Na₂CO₃)

Explanation: Anhydrous sodium carbonate is solid, non-hygroscopic, chemically stable, and meets all criteria for

an ideal primary analytical base standard.

98. Correct Option: a) Nuclear Magnetic Resonance (NMR) Spectroscopy

Explanation: NMR spectroscopy relies on transitions between nuclear spin states in a strong magnetic field to map

carbon-hydrogen skeletal connectivity.

99. Correct Option: a) Precision.

Explanation: Precision measures the degree of mutual agreement and reproducibility within a series of replicated

analytical measurements.

100. Correct Option: a) Purple / Violet.

Explanation: Sulfide radicals react with nitroprusside complexes to form a transient purple/violet coordination

product.

Zoology

101. Correct Option: d) Archaeopteryx

Explanation: The fossil avifauna specimen Archaeopteryx marks a transitional evolutionary bridge, displaying true reptilian jaws lined with teeth.

102. Correct Option: d) Java man

Explanation: Fossil hominid profiles of Java Man (*Homo erectus erectus*) document a characteristic mean cranial

capacity near 900 cc.

103. Correct Option: b) Abiogenesis.

Explanation: Miller's landmark experiment verified organic precursor assembly under primordial conditions, validating chemical abiogenesis theories.

104. Correct Option: b) 25 yrs.

Explanation: A complete adult permanent dental formulation of 2123/2123 requires third molar eruption, signifying an age threshold of adult maturity.

105. Correct Option: b) Ileum.

Explanation: A step-ladder fever configuration alongside trunk rose spots are clinical pathognomonic markers of typhoid, which targets the small intestine's ileum.

106. Correct Option: b) 0.5 micron.

Explanation: The functional thin blood-air respiratory gas exchange membrane across human alveoli measures approximately 0.5 micrometers.

107. Correct Option: a) Diaphragm.

Explanation: The diaphragm is the primary muscle of inspiration driven by the phrenic nerve; its paralysis causes immediate respiratory collapse.

108. Correct Option: b) Coronary artery.

Explanation: The earliest arterial vessels branches emergent from the root segment of the ascending human aorta are the coronary arteries.

109. Correct Option: d) Heart murmur

Explanation: Abnormal internal heart acoustic sounds generated during auscultation by valve structural pathologies are termed murmurs.

110. Correct Option: a) Arbor vitae.

Explanation: The "Arbor Vitae" configuration identifies the highly branched sagittal tree-like appearance of white matter tracts inside the cerebellum.

111. Correct Option: c) Influx of Na⁺ ion

Explanation: The rapid upstroke depolarization phase along action potential paths is driven by the opening of voltage-gated sodium influx channels.

112. Correct Option: b) Urea.

Explanation: Blood leaving hepatic tissue boundaries contains elevated concentrations of urea synthesized by the internal ornithine liver cycle.

113. Correct Option: a) Kangaroo rat.

Explanation: Extreme desert-dwelling kangaroo rats rely on hyper-elongated loops of Henle to maximize hyper-concentrated urinary conservation.

114. Correct Option: b) 280 days.

Explanation: The human gestation period is clinically calculated as 280 days (or 40 weeks) from the first day of the last menstrual period.

115. Correct Option: a) Calcium

Explanation: Calcium ions bind to troponin to trigger a structural shift that unmasks active binding sites on actin filaments during contraction.

116. Correct Option: c) Diabetes insipidus disorder

Explanation: Deficiency of Antidiuretic Hormone (ADH) limits renal water reabsorption, prompting excessive excretion of highly dilute urine.

117. Correct Option: c) Developing spermatozoa

Explanation: Sertoli cells within the seminiferous tubules provide essential structural, regulatory, and nutritive support to developing sperm.

118. Correct Option: c) Opsin and Retinal

Explanation: Rhodopsin is a conjugated photosynthetic visual pigment consisting of the protein opsin and the aldehyde form of vitamin A, retinal.

119. Correct Option: c) Aschelminthes

Explanation: Aschelminthes (roundworms) possess a false body cavity or pseudocoelom that is derived directly from the embryonic blastocoel.

120. Correct Option: c) Choanocytes

Explanation: The central spongocoel cavity of poriferans is lined with flagellated collar cells called choanocytes

that ingest food particles.

121. Correct Option: c) Lack scales and jaws

Explanation: Cyclostomes like lampreys and hagfishes are primitive jawless vertebrates characterized by an absolute absence of scales.

122. Correct Option: c) Cephalochordata

Explanation: In Cephalochordates (e.g., *Amphioxus*), the continuous structural notochord persists throughout life from the head to the tail.

123. Correct Option: c) The liver

Explanation: Earthworm chloragogen cells store nutrients and process metabolic waste products, making them functionally homologous to the human liver.

124. Correct Option: b) Dorsal vessel

Explanation: In the closed circulatory setup of an earthworm, blood flows from the posterior forward exclusively inside the muscular dorsal vessel.

125. Correct Option: a) Bidder's canal

Explanation: In male frogs, mature spermatozoa travel from the vasa efferentia through the kidney matrix to enter Bidder's canal.

126. Correct Option: b) Moving air into and out of the lungs

Explanation: Frogs manipulate the muscular floor of their buccopharyngeal cavity to create positive pressure shifts for lung ventilation.

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127. Correct Option: c) Malignant tertian malaria

Explanation: *Plasmodium falciparum* causes malignant tertian malaria, the most severe clinical form characterized by rapid erythrocyte destruction.

128. Correct Option: c) The patient cannot pass the infection to a mosquito

Explanation: Gametocytes are the sexual stages required to infect vectors; blocking them prevents onward transmission to mosquitoes.

129. Correct Option: b) Reticular connective tissue

Explanation: Reticular connective tissue forms a delicate structural net framework (stroma) that supports cellular elements inside lymphoid organs.

130. Correct Option: c) Histamine, Serotonin, and Heparin

Explanation: Mast cells are specialized elements of connective tissue proper that synthesize and release histamine, serotonin, and heparin.

131. Correct Option: c) Transitional epithelium

Explanation: Transitional epithelium lines the urinary bladder and can accommodate extreme dimensional expansion and stretching without tearing.

132. Correct Option: c) Osteocytes

Explanation: Mature bone cells that are trapped and reside within small fluid-filled lacunae cavities are anatomically termed osteocytes.

133. Correct Option: c) Artificial active immunity

Explanation: Injecting a bacterial toxoid introduces safe antigens that actively induce the host's immune system to synthesize specialized antibodies.

134. Correct Option: c) Helper T-cells (TH)

Explanation: Human Immunodeficiency Virus (HIV) selectively targets and destroys CD4+ Helper T-lymphocytes, collapsing the host immune system.

135. Correct Option: c) Widal test

Explanation: The Widal test is a classical serological agglutination assay utilized in clinical labs to confirm enteric/typhoid fever infections.

136. Correct Option: a) Streptomycin.

Explanation: Streptomycin is an anti-tubercular aminoglycoside antibiotic whose principal adverse toxicity targets the eighth cranial auditory nerve.

137. Correct Option: c) Vitamin A (Beta-carotene)

Explanation: Golden Rice is a bio-fortified agricultural crop engineered to accumulate beta-carotene, a primary metabolic precursor to Vitamin A.

138. Correct Option: b) Calcium in bone has a high atomic number and attenuates X-rays

Explanation: Dense skeletal bone absorbs more X-ray photons due to its highly concentrated calcium content (Z=20), making it appear white.

139. Correct Option: c) Inverted

Explanation: The rapid turnover and lower transient standing biomass of phytoplankton relative to fish consumers yields an inverted aquatic biomass pyramid.

140. Correct Option: c) Endangered

Explanation: According to Red Data Book guidelines, a wild species facing an exceptionally high danger of imminent extinction is classified as Endangered.

BOTANY

141. Correct Option: b) The middle lamella is chiefly made of calcium pectate

Explanation: The middle lamella functions as an intercellular adhesive matrix composed mainly of calcium and magnesium pectate.

142. Correct Option: c) Amyloplasts

Explanation: Amyloplasts are specialized non-pigmented leucoplasts that function exclusively to synthesize and store insoluble starch grains.

143. Correct Option: c) 5g

Explanation: The additive phenotypic range spans 30g across 6 dominant alleles, giving an incremental contribution of 5g per dominant allele.

144. Correct Option: a) 5'-UAGCCUA-3'

Explanation: Anti-parallel RNA transcription base pairing rules convert a template sequence of 3'-ATCGGTA-5' into 5'-UAGCCUA-3'.

145. Correct Option: c) Uracil

Explanation: Uracil is a pyrimidine base structural component unique to RNA molecules, replacing Thymine to pair with Adenine.

146. Correct Option: c) Incomplete dominance

Explanation: Incomplete dominance results in a heterozygous hybrid phenotype that presents as an intermediate blend of both homozygous parents.

147. Correct Option: c) Thomas Hunt Morgan

Explanation: Thomas Hunt Morgan utilized *Drosophila melanogaster** mutants to experimentally establish genetic linkage and verify chromosomal inheritance.

148. Correct Option: c) UAA

Explanation: UAA is a non-sense termination triplet that triggers translation release factors to halt protein synthesis across ribosomes.

149. Correct Option: c) Isobilateral monocotyledonous leaves

Explanation: Isobilateral leaves, which are characteristically found in monocots, display a symmetrical distribution of stomata across both epidermis layers.

150. Correct Option: c) Suberin

Explanation: Outer cork cells synthesized by the phellogen undergo secondary cell wall impregnation with suberin, making them waterproof.

151. Correct Option: c) Exarch

Explanation: Root vascular bundles display an exarch arrangement, where younger metaxylem matures centripetally away from peripheral protoxylem.

152. Correct Option: c) Molecular Oxygen (O₂)

Explanation: Molecular oxygen acts as the final terminal electron acceptor at the end of the mitochondrial ETC, reducing to water.

153. Correct Option: c) Cytokinins
Explanation: The Richmond-Lang effect demonstrates that cytokinins strikingly delay leaf senescence by driving nutrient mobilization.

154. Correct Option: c) Aleurone layer

Explanation: The aleurone layer of germinating seeds synthesizes hydrolytic alpha-amylase in response to embryonic gibberellin signals.

155. Correct Option: a) 2 ATP

Explanation: Splitting one molecule of pre-phosphorylated fructose 1,6-bisphosphate via glycolysis yields a net gain of 2 ATP.

156. Correct Option: c) Ganong's Potometer

Explanation: Ganong's potometer tracks the movement of an isolated air bubble to quantify water absorption rates during plant transpiration.

157. Correct Option: b) Causes constitutive (permanent) activation of the gibberellin signaling pathway

Explanation: Foolish seedling disease symptoms are triggered by a hyperactive mutation that constitutively locks gibberellin signaling pathways open.

158. Correct Option: c) Lysosomes and Peroxisomes

Explanation: Unlike double-membraned nuclei, mitochondria, and plastids, lysosomes and peroxisomes are bounded by a single lipid bilayer.

159. Correct Option: c) Quiescent stage (G₀ phase)

Explanation: Cells failing to pass the G₁ DNA-damage control checkpoint exit active replication to enter a dormant, non-dividing G₀ phase.

160. Correct Option: c) Secondary active symport

Explanation: Sodium-glucose co-transporters utilize secondary active symport, tapping a sodium gradient to actively drive glucose uptake.

161. Correct Option: c) 32

Explanation: A cell with a diploid count of 2n=16 chromosomes carries exactly 32 chromatids (DNA molecules) following S-phase synthesis.

162. Correct Option: c) Nucleolus

Explanation: The nucleolus is a dense nuclear sub-domain dedicated to transcribing ribosomal RNA and assembling immature ribosomal sub-units.

163. Correct Option: c) Pyrenoid

Explanation: Algal pyrenoids are chloroplast micro-compartments featuring a RuBisCO-rich protein core surrounded by a starch sheath.

164. Correct Option: c) Protonema

Explanation: Germinating haploid moss spores initially form a prostrate, filamentous, green transitional stage called a protonema.

165. Correct Option: c) Strobili or Cones

Explanation: In the horsetail pteridophyte *Equisetum**, sporangia are organized into compact, specialized terminal clusters termed strobili.

166. Correct Option: c) Anemophily (Wind pollination)

Explanation: The wing-like saccate air bladders on *Pinus** pollen grains lower particle density to optimize dispersal via wind (anemophily).

167. Correct Option: c) 7-celled, 8-nucleate

Explanation: Standard angiosperm megasporogenesis and subsequent mitotic divisions produce a mature female gametophyte containing 7 cells and 8 nuclei.

168. Correct Option: c) A typical pteridophyte

Explanation: Pteridophytes (ferns) feature an independent alternation of generations, where both sporophyte and gametophyte are photosynthetic and free-living.

169. Correct Option: a) Mustard family (Brassicaceae)

Explanation: A tetradynamous androecium consisting of 4 long inner and 2 short outer stamens is a diagnostic family trait of Brassicaceae.

170. Correct Option: c) Prions

Explanation: Spongiform encephalopathies like Mad Cow Disease are caused by prions, which are slow-acting infectious agents made purely of protein.

171. Correct Option: c) Only the viral nucleic acid genome

Explanation: During infection of *E. coli**, a T-even bacteriophage leaves its capsid shell outside and injects only its naked nucleic acid core.

172. Correct Option: a) $\log S = \log C + Z \log A$

Explanation: Humboldt's species-area power function $S = CA^Z$ linearizes on a log-log coordinate grid as $\log S = \log C + Z \log A$.

173. Correct Option: c) Proto-cooperation (Facultative Mutualism)

Explanation: The hermit crab-sea anemone association is proto-cooperation; both organisms derive clear mutual benefits but can survive independently.

174. Correct Option: c) Lichens

Explanation: Lichens are the primary pioneer colonizers on bare rock substrates, initiating ecological succession by weathering solid minerals.

175. Correct Option: c) 100J

Explanation: Applying Lindeman's 10% thermodynamic efficiency rule: 10,000J at producers drops to 1,000J at primary consumers, and 100J at secondary consumers.

176. Correct Option: c) Apical and axillary meristems

Explanation: Actively dividing shoot apical and axillary meristems lack vascular connection lines, rendering them free of viruses and ideal for clonal micropropagation.

177. Correct Option: c) Solanaceae

Explanation: Atropine is a secondary anticholinergic alkaloid commercially isolated from *Atropa belladonna**, a member of the Solanaceae family.

178. Correct Option: c) Polyethylene glycol (PEG) or Electrofusion

Explanation: Stripped somatic protoplasts are chemically coerced to fuse using polyethylene glycol (PEG) or high-voltage electrofusion fields to bypass species barriers.

179. Correct Option: a) Molecular scissors.

Explanation: Restriction endonucleases are called molecular scissors due to their ability to cleave internal DNA phosphodiester bonds at specific sites.

180. Correct Option: b) *Agrobacterium tumefaciens*.

Explanation: The tumor-inducing Ti plasmid, extensively utilized as a transgenic plant cloning vector, is native to

*Agrobacterium tumefaciens**

MAT -

181. Correct Option: b) NKDULGUD

Explanation: The cryptographic cipher alters positional letters forward and backward alternately, encoding KHARIDAR into NKDULGUD.

182. Correct Option: c) 2, 1, 5, 4, 3

Explanation: Sorting by increasing geographic scale yields: Ward (2), District (1), Municipality (5), Province (4), and Country (3).

183. Correct Option: a) Desk

Explanation: A pen is filled with ink, and since the substitution cipher dictates that 'Ink' is called 'Desk', the student fills it from the 'Desk'.

184. Correct Option: b) Grandson.

Explanation: The sister of a man's son is the man's daughter; if she is the woman's mother-in-law, the woman's husband is the man's grandson.

185. Correct Option: c) H8S19

Explanation: Alphanumeric pairs add up to 27 based on alphabet position indices, making the localized paired index group H8S19.

186. Correct Option: a) 65.

Explanation: The series follows a second-order logic of adding doubling values: $3 (+2) \rightarrow 5 (+4) \rightarrow 9 (+8) \rightarrow 17 (+16) \rightarrow 33 (+32) \rightarrow 65$.

187. Correct Option: a) Rs. 90

Explanation: Setting up fractional cost equations $1.25(0.8x) = 1.2x - 18$ yields a calculated initial cost price of exactly Rs. 90.

188. Correct Option: a) 49 years.

Explanation: The total age increment is 16 years, making the combined age of the new trainees 98 years and their

mean age 49 years.

189. Correct Option: a) 24 hours.

Explanation: Combined hourly drainage work capacity parameters evaluate to: {24}, requiring 24 hours to fill the tank.

$\frac{1}{6} - \frac{1}{8} = \frac{1}{24}$

190. Correct Option: a) 7.2 days.

Explanation: Work rate equation balance precisely to 7.2 days.

$\frac{1}{10} + \frac{1}{15} = \frac{1}{5d} = 1$ simplifies to $5d = 36$, which resolves

191. Correct Option: a) South-East.

Explanation: Tracking positional vector translations (10m East, 5m South, 10m East) locates the individual south and east of their starting origin.

192. Correct Option: a) 22nd.

Explanation: Positional ranking from the bottom of the row is evaluated as: $35 - 14 + 1 = 22$.

193. Correct Option: c) R120N

Explanation: Letters reverse while numerical values grow by changing factors, generating the sequential entry solution R120N.

194. Correct Option: a) 15 km West.

Explanation: The translation forms an open rectangle (30km N, 15km W, 30km S), leaving the trekker exactly 15km West of the start coordinates.

195. Correct Option: a) West

Explanation: At 3:45 PM, the clock's minute hand points precisely at the 9 o'clock index line, which corresponds to West.

196. c

197. b

198. d

199. c

200. b